

3. Calcule o equivalente de Thévenin entre os pontos  $a$ - $b$  da Figura 3.

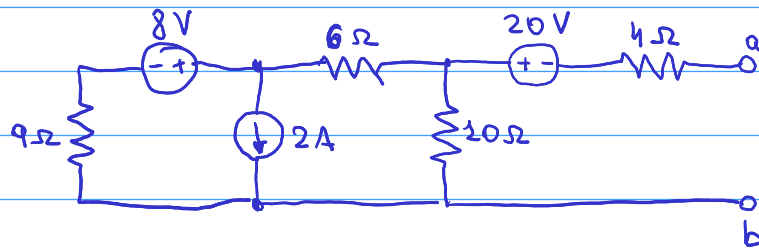
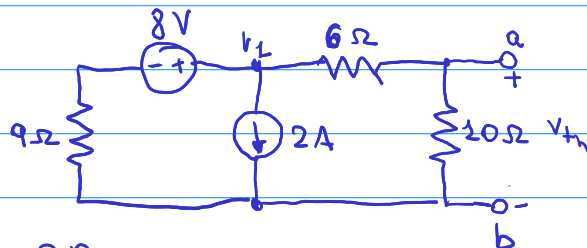


Figura 3

Calculando  $V_{th}$



$$V_{th} = \frac{10}{16} \cdot v_1 - 20$$

$$\frac{v_1 - 8}{9} + 2 + \frac{v_1}{16} = 0$$

$$\frac{288}{160}$$

$$\frac{64}{64} \frac{16}{4}$$

$$16v_1 - 128 + 288 + 9v_1 = 0$$

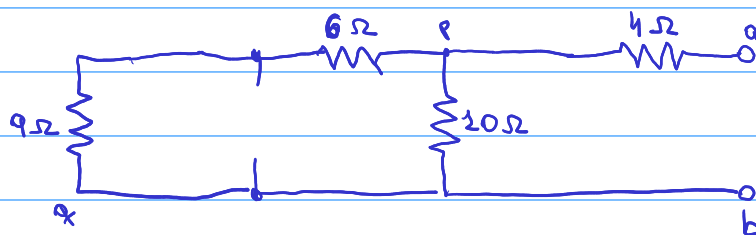
$$\frac{160}{150} \frac{25}{6,4}$$

$$25v_1 = -160$$

$$v_1 = -6,4 \text{ V}$$

$$V_{th} = \frac{10 \cdot (-6,4)}{16} - 20 = -24 \text{ V}$$

$$\frac{150}{150} \frac{25}{6}$$



$$R_{eq} = 4 + (6+9) \parallel 10 = 4 + \frac{15 \cdot 10}{25} = 10 \Omega = R_{th}$$

